

$$= -3.00 \times 5.00 + \frac{1}{2} 9.81(5.00)^2 = 108 \text{ m}$$

- 5 C** By conservation of momentum:
 $(2.0)(0.5) = (2.0 + 0.500)v_f \rightarrow v_f = 0.4 \text{ m s}^{-1}$.
 change in momentum = $p_f - p_i = 2.0 (0.4 - 0.5) = -0.2 \text{ N s}$

- 6 B** By ΔP , upthrust = weight of fluid displaced. Liquid X has smaller density and hence